

THE CAPSTONE BLUEPRINT

From Messy Ideas to a Mature and Feasible Proposal

OSALP 2025-04 | Funded by the Gifted Education Fund

Department of Statistics and Data Science
The Chinese University of Hong Kong

PROGRAMME ACKNOWLEDGEMENT AND DISCLAIMER

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PROGRAMME CONTEXT Prepared for the Online Summer Advanced Learning Programme in Statistics and Data Science (OSALP 2025-04), organised by the Department of Statistics, The Chinese University of Hong Kong.

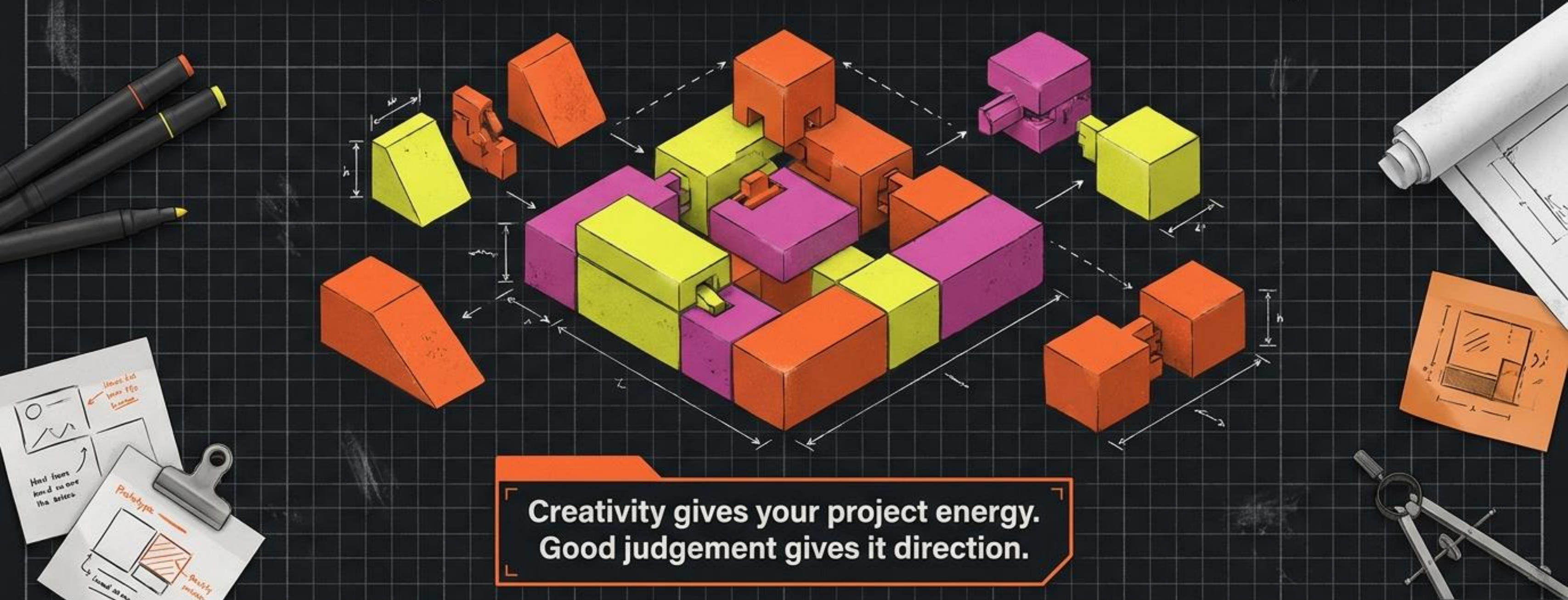
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THE CAPSTONE BLUEPRINT

From Messy Ideas to a Mature & Feasible Proposal



**Creativity gives your project energy.
Good judgement gives it direction.**

The Real Goal of Your Proposal

Your first proposal doesn't need to be perfect, or sound like a university research paper. It's a starting point to question, test, and improve.

The Idea



The Mature Proposal

The Ultimate Target

- ✓ Can 5 secondary students complete it?
- ✓ Can you understand and defend it?
- ✓ Can you present it confidently in 20 minutes?

A mature proposal makes thoughtful choices about what to keep, what to simplify, and what to leave as an extension.

Choose Your Pathway

Pathway 1: Fund Risk Analysis

Focus: Analyse or compare investment funds.

Must consider: Real price/return data, return vs volatility, drawdown/downside loss, Value at Risk (VaR), correlation/diversification, normal vs stress scenarios, simulations, and a conclusion for a specific investor type.

Warning: Don't just find the highest historical return. What do risk & return mean for the investor?

Pathway 2: Insurance or Annuity Innovation

Focus: Design an innovative product supported by real data & simulation.

Must consider: The financial loss, target customer, claim frequency/severity, expected loss/premium logic, pooling, waiting periods/deductibles, exclusions, normal/stress scenarios, affordability & sustainability.

Do not add a method only because it sounds advanced.
Use it because it helps answer an important question.

Titles Are Not Purposes

Flashy Titles

~~Crypto Wallet Insurance~~

~~AI Job-loss Insurance~~

~~Bitcoin vs. Treasury ETF~~

~~Bitcoin v Insurance~~

Interesting, but what will you actually do?

Guiding Question Generator

The Solution

Complete this sentence:

We hope to analyse, design or compare...

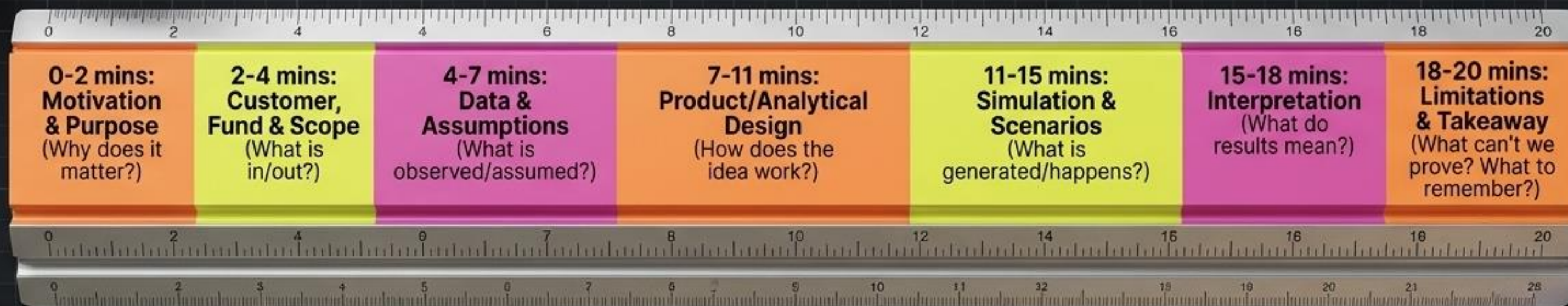
Fund Project Guiding Question:

How do the selected funds compare after considering both return and downside risk?

Insurance Project Guiding Question:

Can a simple product protect the stated financial loss at a premium that may be affordable and sustainable?

The 20-Minute Endgame



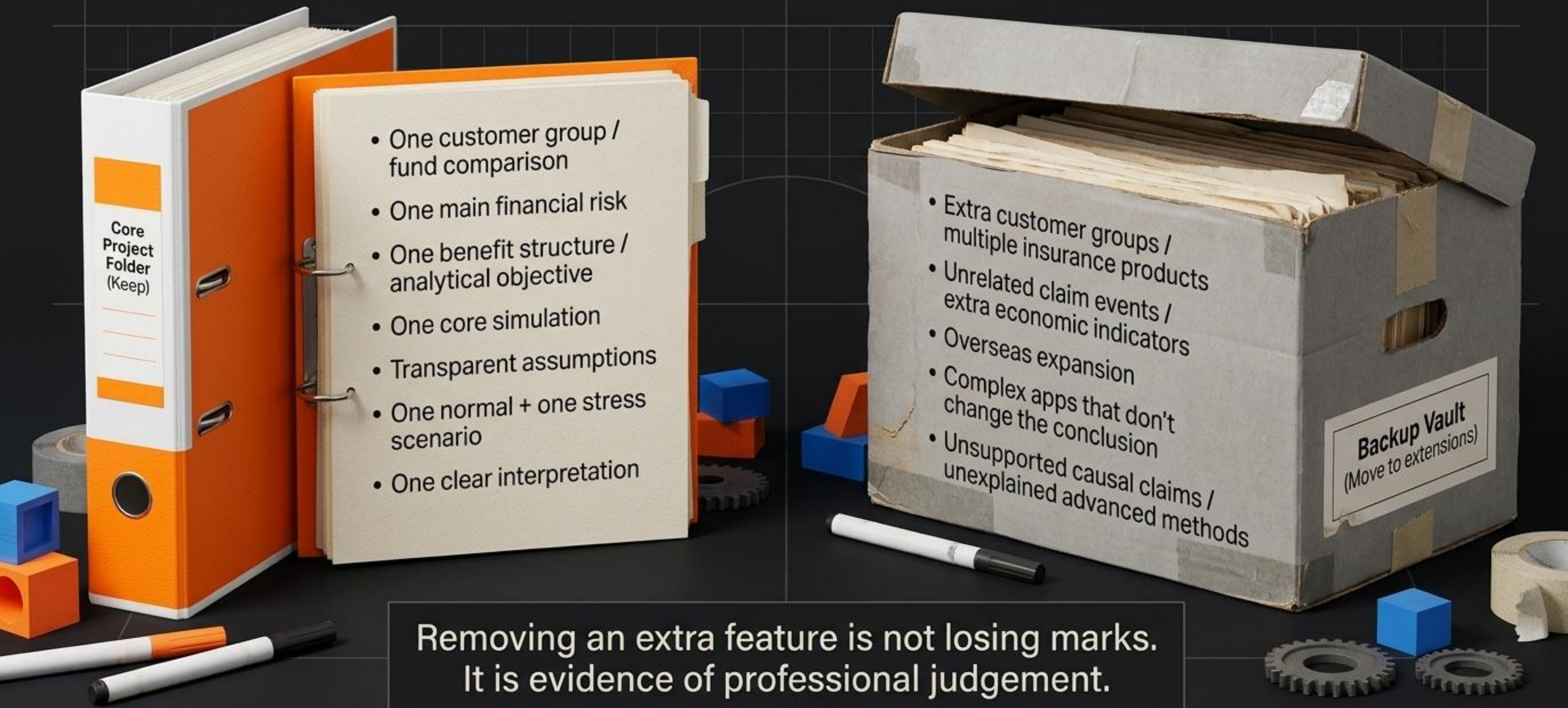
☐ Explain every tech method in 1 minute?

☐ One clear story?

☐ One honest limitation?

☐ One memorable takeaway?

If We Cut Half the Proposal, What Survives?



Be Specific. “Students” or “Gig Workers” is Not Enough.

Diagnostic Questions

- Who experiences the loss?
- Who buys the product?
- Who pays the premium?
- Who submits the claim?
- Who receives the benefit?
- Who verifies the evidence?

Fund Note: Identify the decision-maker (e.g., cautious, growth-oriented, or loss-averse investor).

“Elevator Pitch” Mad-Libs

Our target customers are [____], who currently manage this risk by [____]. Our product may appeal to them because it offers [____], which the existing alternative does not offer.

The Reality Check: What Else Could Work?



Diagnostic Checklist

- ☒ What simpler solution could manage the same risk?
- ☒ What financial loss remains after existing support?
- ☒ Is the loss severe/uncertain enough to justify insurance?

The Mature Conclusion: Small and predictable costs are handled through direct payment, while insurance protects against the less frequent but more severe part of the loss.

An Unfortunate Event is Not Automatically an Insurable Loss

My phone broke

I failed my test

I got fired

Market crashed

Verified repair or replacement cost

Examination re-entry fee

Temporary income interruption / Reskilling cost

Portfolio drawdown / Loss above a stated threshold

Insurance

A claim occurs when [.....].

Required evidence is [.....].

It pays [.....] up to [.....].

Fund

We analyse [.....] over [.....].

Return is measured using [.....].

Downside risk is [.....].

The 3-Number Story

If you can't explain what happens to ONE representative customer/investor, it's too complicated.



Insurance Example Checklist

- ✓ Premium paid
- ✓ Covered event / Evidence required
- ✓ Loss amount
- ✓ Deductible / Co-payment
- ✓ Benefit paid & Maximum limit



Fund Example Checklist

- ✓ Starting investment
- ✓ Return or loss
- ✓ Effect of volatility/leverage
- ✓ Relevant risk measure
- ✓ Meaning for intended investor

Can we explain the central idea using no more than three numbers?

Data Needs a Job Description



What is measured? (Variable, unit, population)



Where is it from? (Original source & period available)



Why is it relevant? (Which input does it support: frequency, severity, return?)



Limitations? (Missing values, short history)

Notes: _____

- Labor

[Observed
Data]

[Derived
Estimate]

[Assumption]

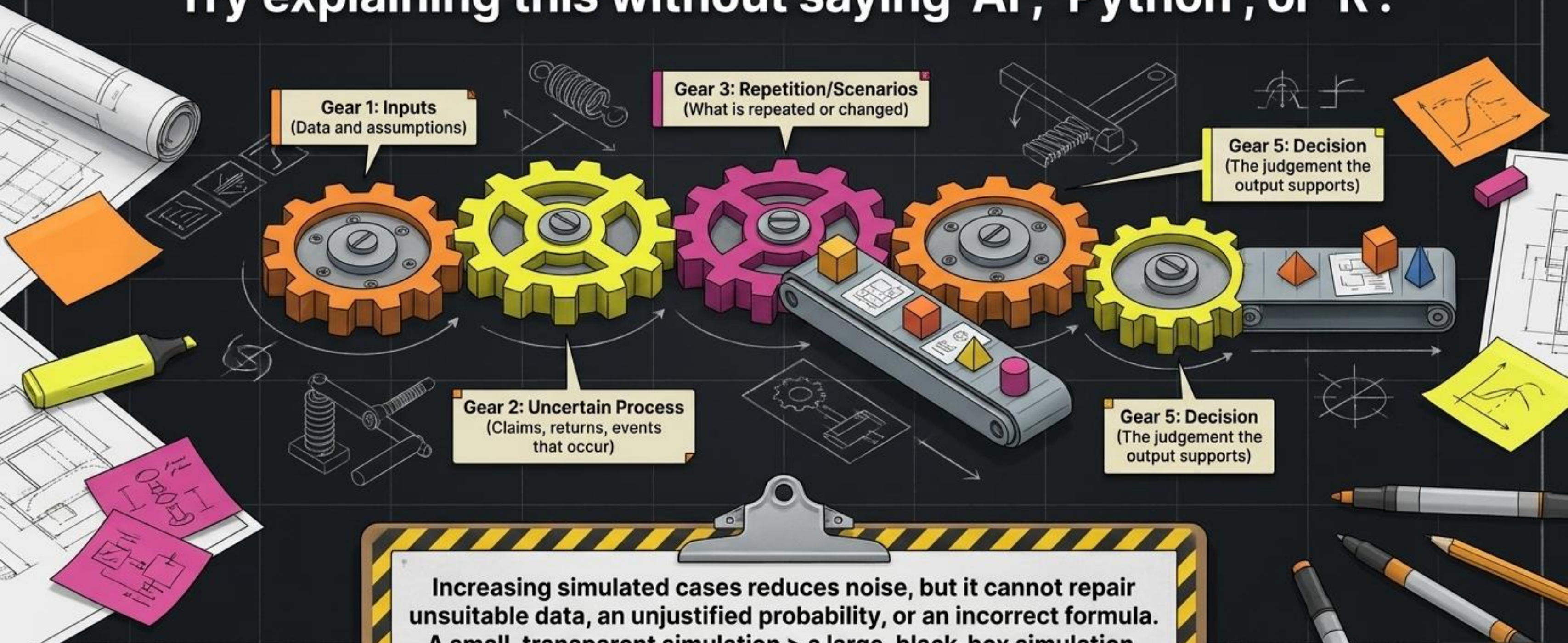
[Scenario
Choice]

Be especially careful NOT to:

- Use international data as a Hong Kong probability.
- Treat AI-task exposure as job-loss probability.
- Treat correlation as proof of causation.
- Allow AI to fill a missing number without a source.

The 5-Gear Simulation Model

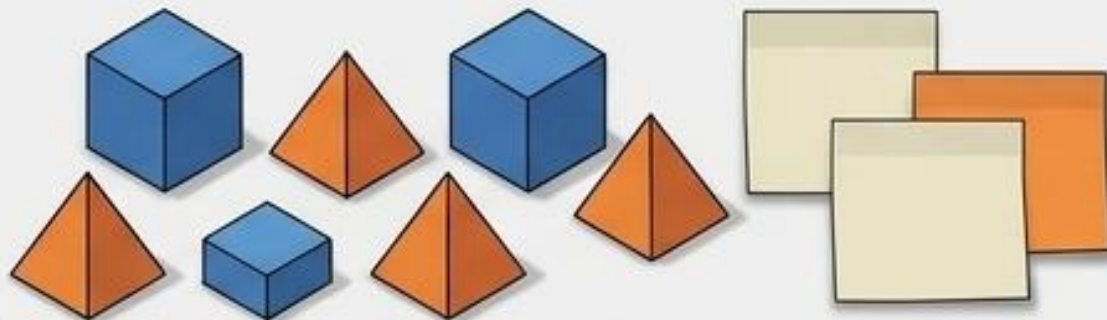
Try explaining this without saying 'AI', 'Python', or 'R'.



What Happens When Things Go Wrong?



Normal Conditions



Stressful Conditions

Fund Stress Variables:

Change volatility, correlation, interest rates, liquidity, drawdown.

Insurance Stress Variables:

Change claim frequency, severity, duration, benefit level, simultaneous claims.

Trigger Events:

Cyberattacks, Mass Layoffs, Pandemics, Transport Disruption, Market Crashes.

State exactly which assumptions change in your stress scenario

A Fallback is Not Failure. It is Project-Risk Management.

Real projects break.
Datasets vanish. Apps fail.
Advanced methods become
too difficult.

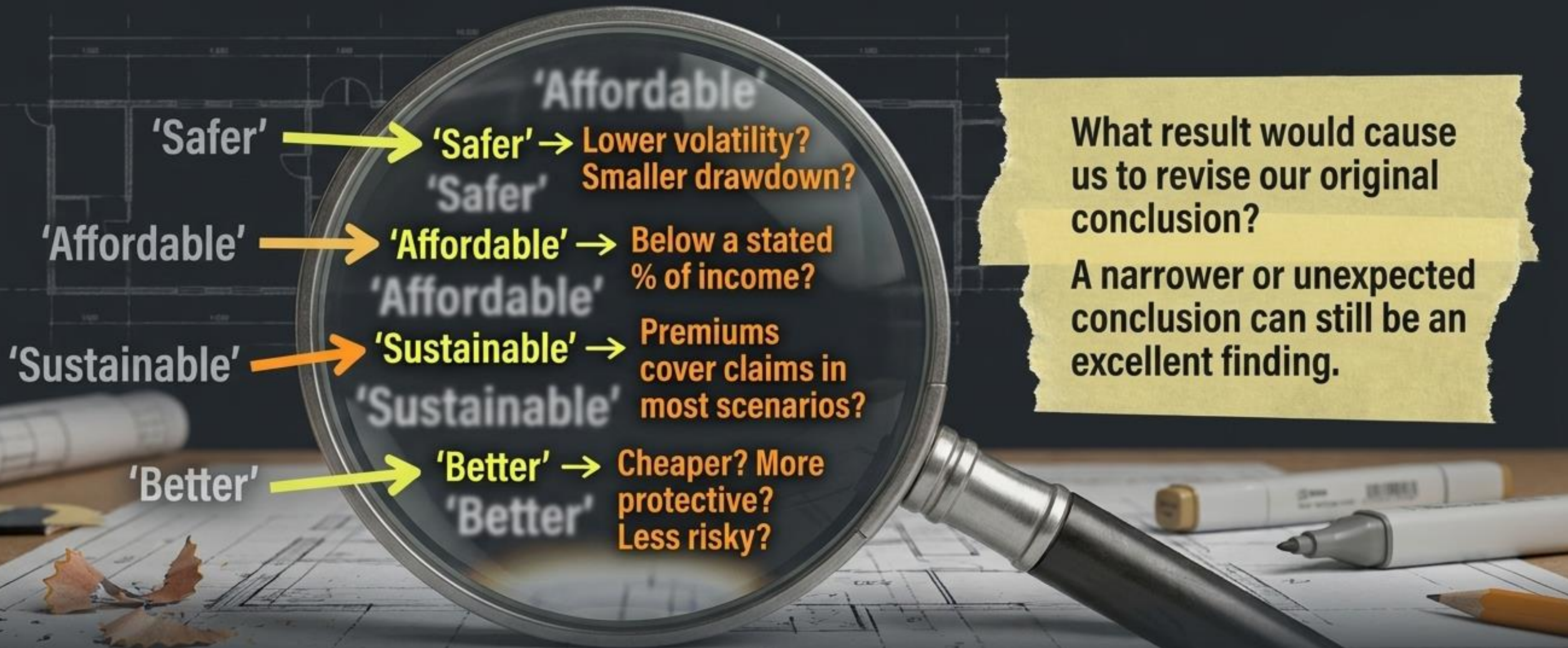
What is your simpler version?

**BREAK
IN CASE OF
EMERGENCY**

Fallback Toolkit

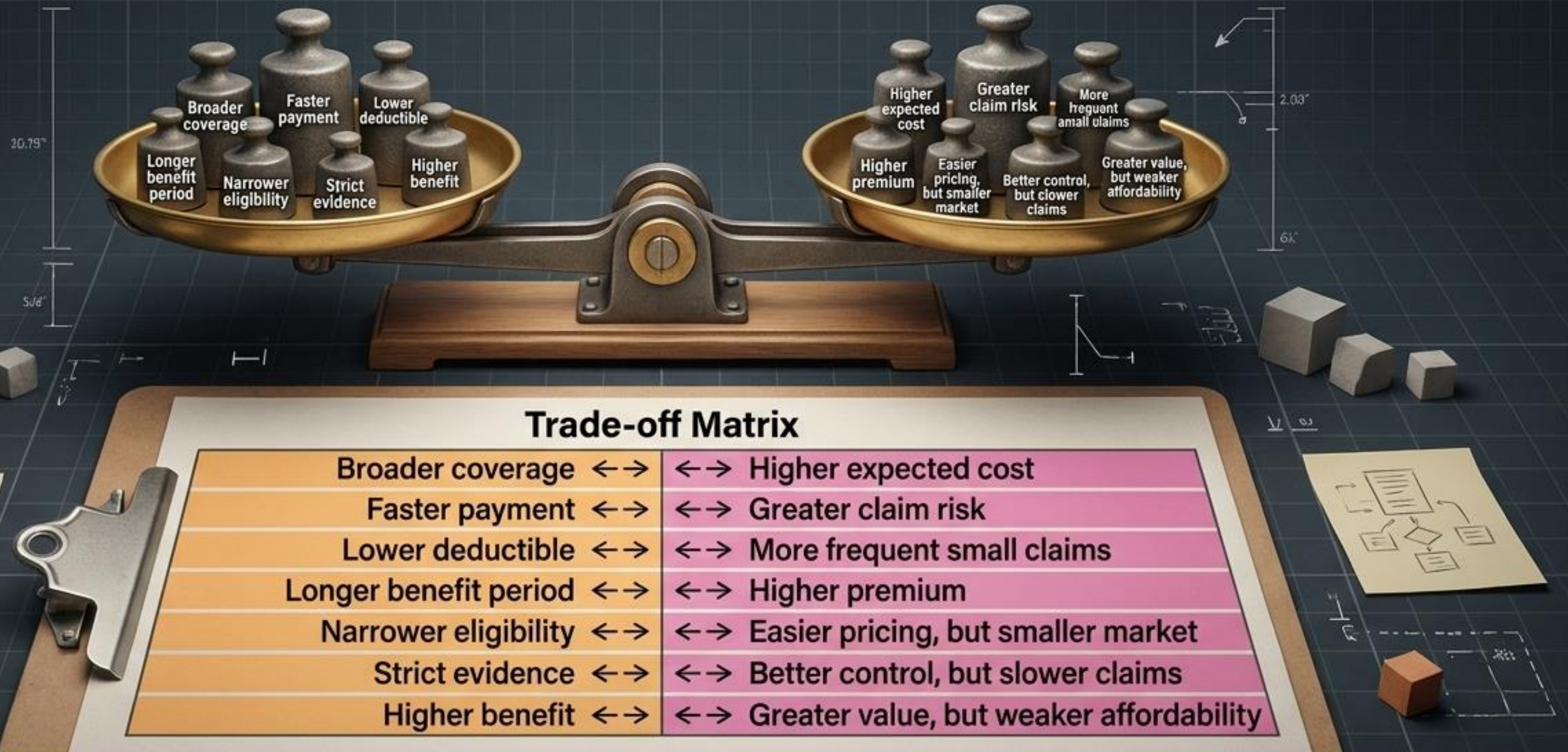
- ☒ Reduce the customer group or number of funds.
- ☒ Use fewer variables.
- ☒ Replace market-price estimates with transparent scenarios.
- ☒ Use a discrete probability table.
- ☒ Use one stress test instead of complex optimization.
- ☒ State the missing data as a clear limitation.
- ☒ Remove a conclusion you cannot support.

Define Your Buzzwords



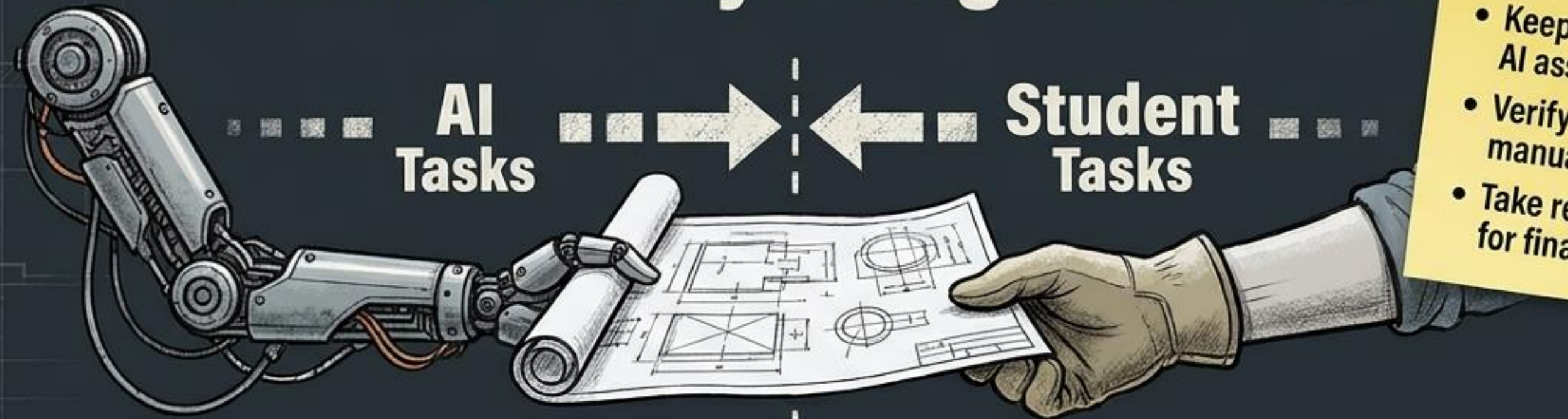
Warning: Be careful with **absolute** words like *prove, superior, safest, optimal*.

There is No Perfect Product, Only Choices



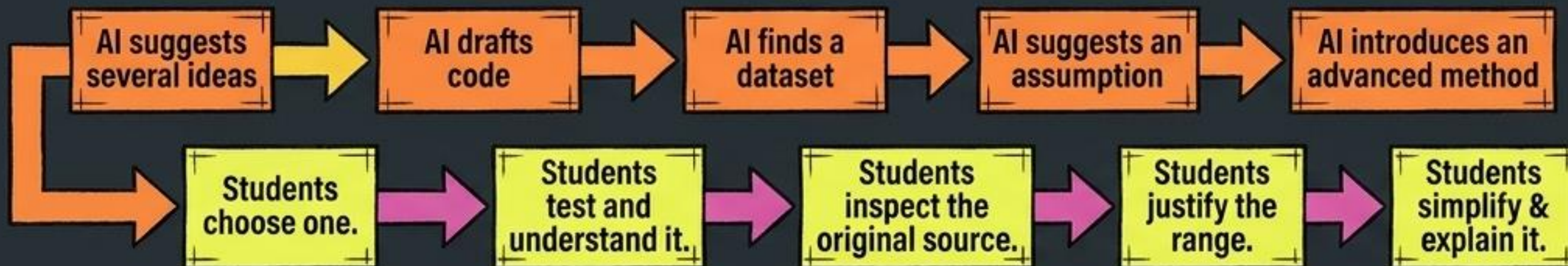
Lowering the premium without changing the risk will not solve the problem!

Who is Actually Doing the Work?



Mandatory Rules

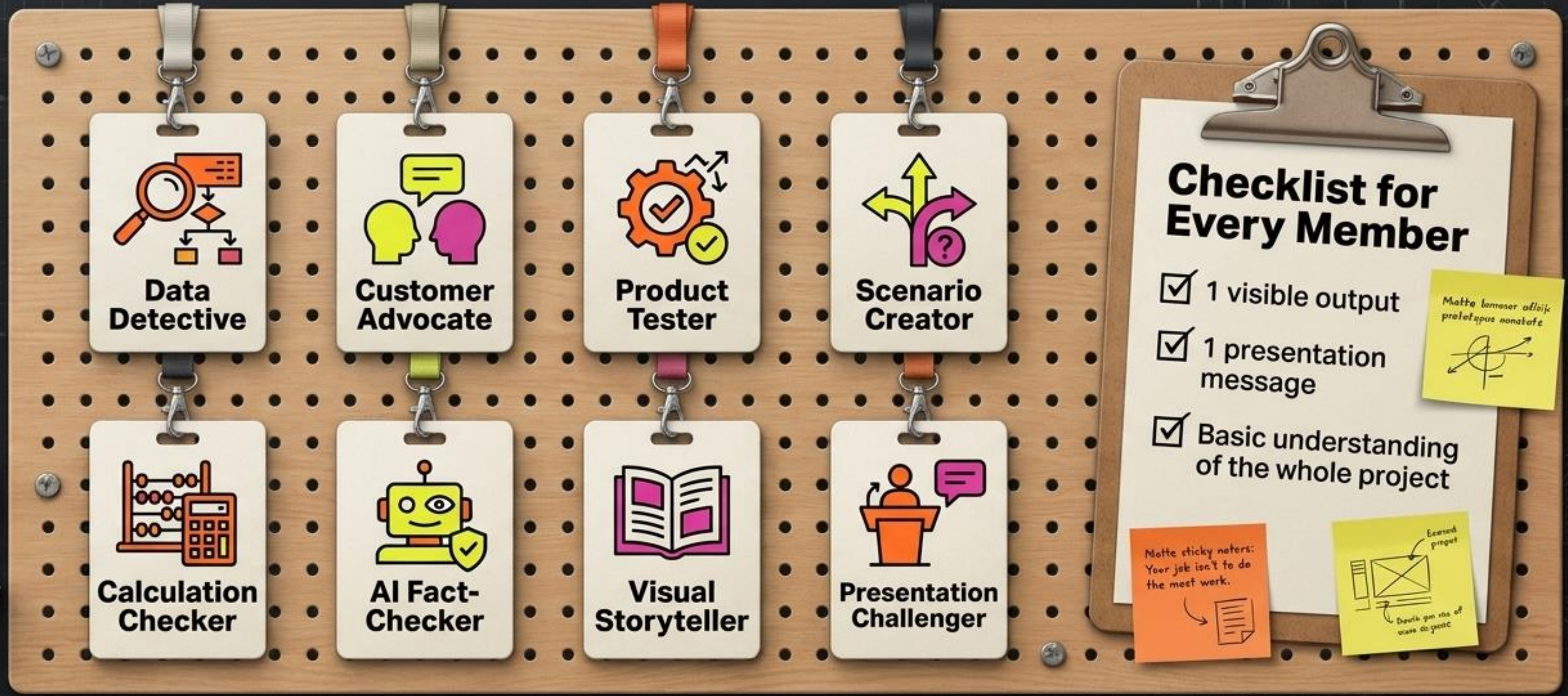
- Keep a record of AI assistance.
- Verify small examples manually.
- Take responsibility for final meaning.



The Golden Rule

The key question isn't whether AI was used. **It's whether you transformed AI assistance into student-owned understanding.**

Choose Your Class (No Blank Roles)



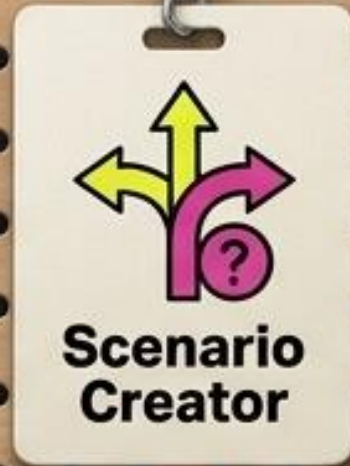
**Data
Detective**



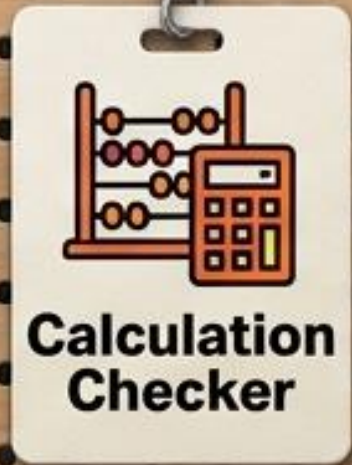
**Customer
Advocate**



**Product
Tester**



**Scenario
Creator**



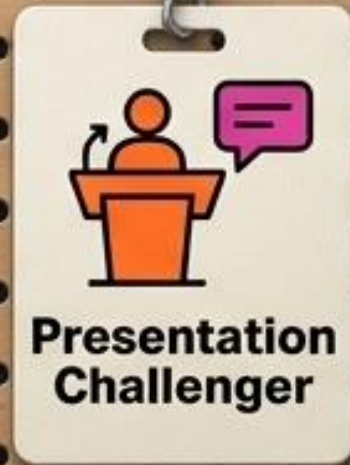
**Calculation
Checker**



**AI Fact-
Checker**



**Visual
Storyteller**



**Presentation
Challenger**



Checklist for Every Member

- ☒ 1 visible output
- ☒ 1 presentation message
- ☒ Basic understanding of the whole project

Motte learner of the project



Motte sticky notes:
Your job isn't to do
the most work.



Note to Coordinators: Your job isn't to do the most work.
It's to help the group make decisions and include everyone.

The Consultation Boss-Fight: Are You Ready?

What is the main thing you will analyse, design, or compare? ?

What is essential vs. what is an extension?

Who is the exact target customer/investor? ?

What simpler alternative could solve this?

Which exact dataset has the numbers you need? ?

What happens in ONE simulated case?

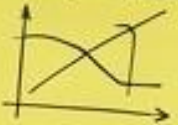
Which input is observed vs. assumed? ?

What could reverse the result?

What can the project NOT prove? ?

What will each member complete next?

Matte lenner ofikis
prototypes nonotete



These questions aren't traps. They help you find the strongest version of your project.

System Check: Is Your Proposal Mature?

- ☒ One clear purpose
- ☒ Core explained in 20 minutes
- ☒ Target customer/investor is specific
- ☒ Problem is meaningful + alternatives considered
- ☒ Datasets confirmed & assumptions labeled
- ☒ One numerical example can be explained
- ☒ Simulation has clear inputs, events, outputs
- ☒ Normal & stress scenarios included
- ☒ Simpler fallback plan ready
- ☒ Conclusion not decided in advance
- ☒ AI work verified manually
- ☒ Every member has a visible contribution
- ☒ Audience leaves with one memorable takeaway

The Ultimate Blueprint Template

Purpose: We hope to...

Target Customer: ...

Main Risk: ...

Observed Data: ...

3-Number Example: ...

Normal Scenario: ...

Fallback Plan: If [X] fails, we will...

Core: Project works even if we remove...

Current Alternative: ...

Most Important Dataset: ...

Assumptions: ...

Sim Inputs/Outputs: ...

Stress Scenario: ...

Main Limitation: ...

20-Minute Takeaway: The audience should remember...

A mature and feasible proposal has a clear purpose, a manageable scope, a realistic customer, transparent assumptions, an explainable simulation, a practical fallback, an honest limitation, and visible ownership from every group member.

BUILD IT CLEARLY.
TEST IT HONESTLY.
EXPLAIN IT YOURSELF.

A mature proposal is focused, feasible, transparent and student-owned.

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